

Project Report

Data Analysis Using Web Scraping

Course: SDS 106 – Algorithms and Programming 2

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Introduction

In this project, a comprehensive data analysis of anime content was conducted using data collected from the website World-Art.ru. The dataset includes a wide range of attributes such as ratings, number of episodes, genres, release dates, and descriptive information about each anime title.

The primary objective of this study is to identify meaningful patterns and trends within the dataset and to better understand how different characteristics of anime influence audience ratings and overall perception. By applying data analysis techniques, the project aims to transform raw scraped data into structured insights.

This project also demonstrates the importance of combining technical skills, such as web scraping and data preprocessing, with analytical thinking in order to extract value from real-world datasets.

Motivation

The motivation behind this project extends beyond fulfilling academic requirements. It is driven by the need to gain hands-on experience in working with real-world data, which is often incomplete, unstructured, and inconsistent.

Anime is a rapidly expanding global industry with a diverse audience. Understanding trends in this domain can provide valuable insights into content preferences, production strategies, and audience behavior.

Moreover, this project allows the development of practical skills in data collection, cleaning, feature engineering, and visualization. These skills are essential for any data-driven profession and are widely applicable across various industries.

Data Source Description

The dataset used in this project was collected from the website:

http://www.world-art.ru/animation/list.php?limit_1=0

This resource provides structured information about anime titles across different years and genres. The final dataset contains 5826 records, making it sufficiently large for statistical analysis.

Each record represents an individual anime entry and includes multiple attributes such as title, rating, number of episodes, genres, release date, duration, and additional descriptive fields.

The dataset offers a comprehensive overview of anime production over time and serves as a reliable foundation for exploratory data analysis.

Scraping Methodology

The data collection process was implemented using Python and several specialized libraries. The main tools included requests for sending HTTP requests, BeautifulSoup for parsing HTML content, and pandas for storing and processing structured data.

The website uses pagination controlled by the parameter limit_1, which increases in steps of 25. A loop was designed to iterate through all available pages and collect data sequentially.

Special attention was given to encoding, as the website uses windows-1251 format. Proper handling of encoding ensured correct extraction of text data.

The collected data was then stored in a CSV file, allowing further processing and analysis.

Data Cleaning

Data cleaning is a critical step in any data analysis project. The raw dataset contained missing values, inconsistencies, and potential errors that needed to be addressed.

Missing values in key columns such as rating and release date were removed, while other missing values were filled using appropriate strategies (e.g., mode or default values).

Data types were standardized, duplicates were removed, and outliers were identified using statistical methods such as the Z-score technique.

These steps ensured that the dataset became reliable and suitable for further analysis.

Exploratory Data Analysis (EDA)

Exploratory Data Analysis (EDA) was performed to better understand the structure, distribution, and relationships within the dataset before applying deeper analysis.

First, a general overview of the dataset was conducted, including checking the number of records, data types, and missing values. This step helped to identify inconsistencies and determine necessary preprocessing actions.

Next, the distribution of key variables was analyzed. In particular, the rating variable was examined using histograms. The results showed that ratings are concentrated within a relatively narrow range, indicating that most anime receive moderate to high evaluations.

The number of episodes was also analyzed. It was observed that many anime have multiple episodes, confirming that serialized formats dominate over standalone movies.

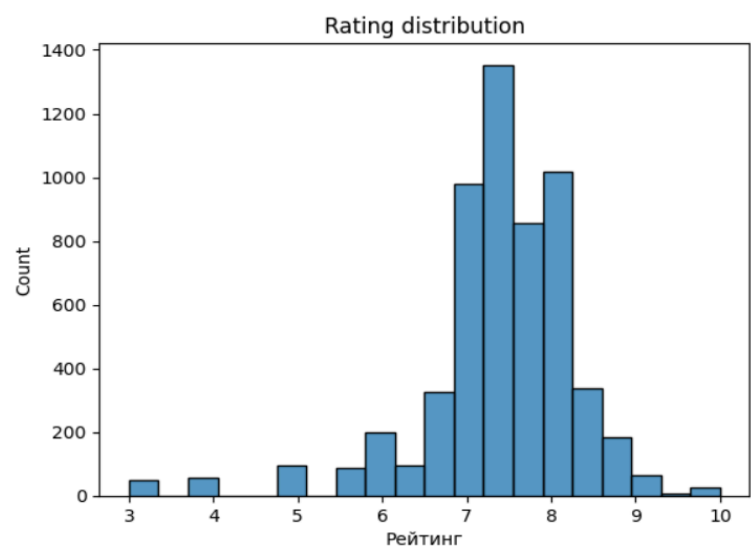
Temporal analysis was performed using the release date variable. By grouping data into decades, it became clear that anime production has increased significantly over time, especially after the 2000s.

Categorical variables such as genres were also explored. Some genres appeared more frequently than others, suggesting dominant themes within the dataset.

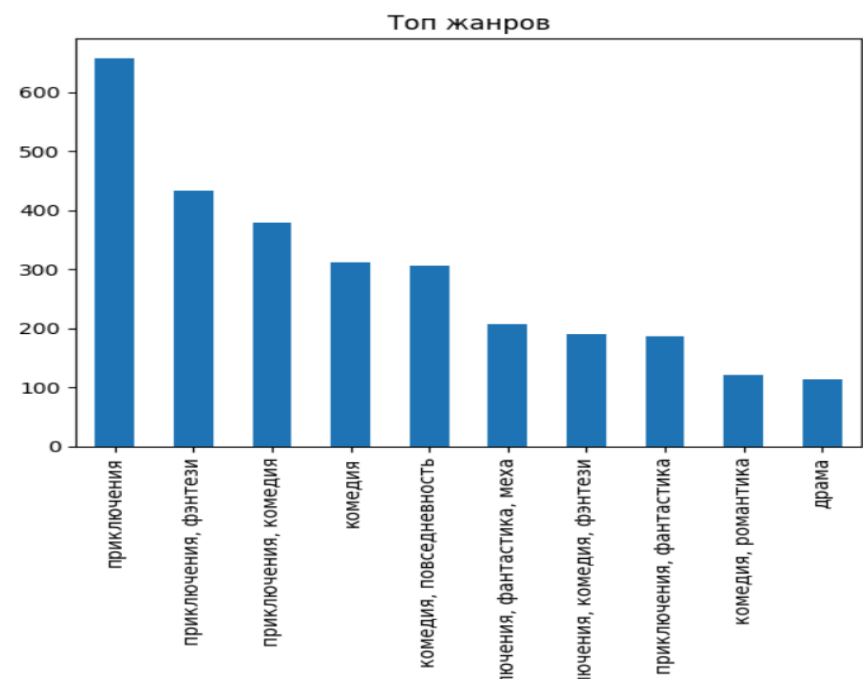
Additionally, boxplots were used to identify outliers in numerical variables. While some extreme values were detected, they did not significantly affect the overall distribution.

EDA played a crucial role in understanding the dataset and guiding further analysis.

Data Visualization



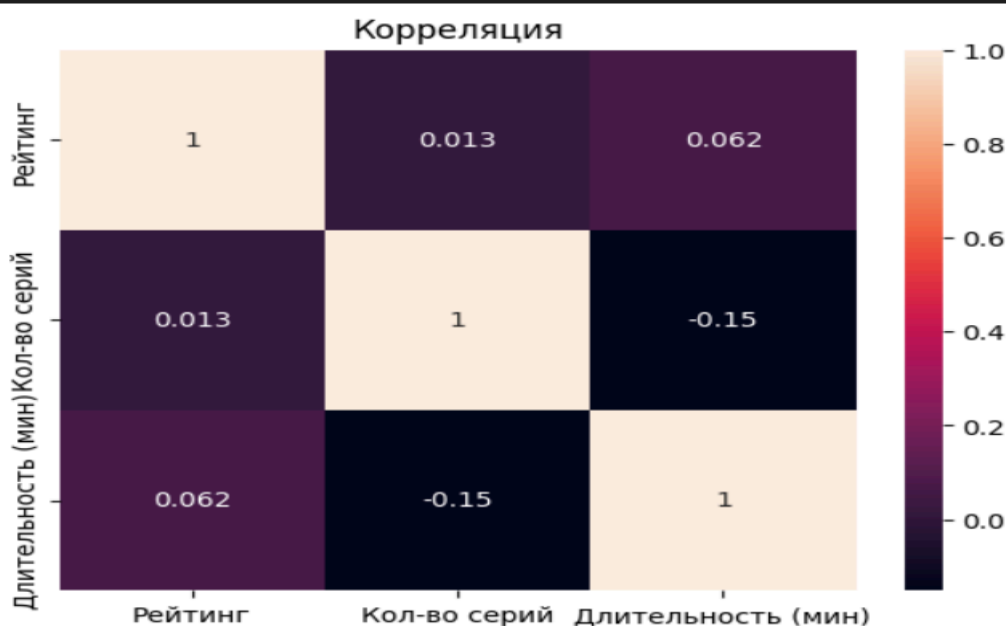
This histogram shows the distribution of anime ratings. Most values are concentrated between 6.5 and 8.5, indicating that the majority of anime receive moderately high evaluations. Extremely low and extremely high ratings are relatively rare, which suggests that audience perception tends to be consistent and not highly polarized.



This bar chart illustrates the most common genres in the dataset. Adventure-related genres dominate, followed by combinations such as adventure-fantasy and adventure-comedy. This suggests that viewers prefer dynamic and engaging storytelling with elements of action and imagination.



The scatter plot represents the relationship between duration and rating. The data points are widely scattered, indicating that there is no strong correlation between duration and rating. Both short and long anime can receive high or low ratings, suggesting that duration is not a key factor influencing audience evaluation.



The heatmap shows correlations between numerical variables such as rating, number of episodes, and duration. The correlation values are generally low, indicating weak relationships between variables. This suggests that ratings are influenced by multiple factors rather than a single dominant variable.



This line chart shows the average rating over time. Despite fluctuations, the overall trend remains relatively stable. This suggests that audience evaluation standards have remained consistent over the years, even as the industry has evolved.

Analysis

The analysis of the dataset was conducted using both statistical methods and exploratory data analysis techniques. The primary focus was on identifying patterns in ratings, production trends over time, and structural characteristics of anime content.

First, the distribution of ratings was examined. It was observed that most values are concentrated within a relatively narrow range, indicating that audience evaluations tend to be consistent and rarely extreme. This suggests that the rating system reflects a stable perception of quality rather than highly polarized opinions.

Secondly, temporal analysis was performed by grouping data into decades. The results show a noticeable increase in the number of anime releases starting from the early 2000s. This trend reflects the rapid growth of the anime industry, likely driven by globalization, digital platforms, and increased accessibility.

Another important aspect of the analysis was the comparison between movies and serials. The data clearly indicates that the majority of anime productions are serialized rather than standalone films. This highlights the dominance of episodic storytelling in the industry.

Additionally, outlier detection was performed using statistical methods. While some extreme values were identified, they did not significantly distort the overall trends, which confirms the robustness of the dataset.

KEY FINDINGS & INSIGHTS

Insights

Based on the analysis, several key insights can be identified:

- **The majority of anime ratings fall within a stable and narrow range, suggesting consistent audience evaluation behavior**
- **Serial anime dominates over movies, indicating industry preference for long-form storytelling**
- **There has been a significant increase in anime production after the 2000s, reflecting industry expansion**
- **High-rated anime are present across all decades, which suggests that quality content is not limited to a specific time period**
- **Structural characteristics such as number of episodes and genre may influence ratings, although further analysis is required**

These insights demonstrate that while the anime industry continues to grow, core audience preferences remain relatively stable.

BONUS TASK:

This script builds a customized recommendation engine using a Random Forest model.

It works by first establishing a "quality threshold"—automatically marking any anime

with a rating above 8.5 as a positive preference. This serves as the target for

the model to learn what makes a show successful in this dataset.

To handle the descriptive data, the code breaks down the "Genres" string into a

binary matrix where each unique genre becomes its own column. It also converts

the "Original_type" into dummy variables and includes technical specs like

episode counts and duration as numerical features.

Once the data is prepared, the Random Forest algorithm analyzes the relationship

between these features and the high-rating label. After training, the model doesn't

just look at existing ratings; it calculates a "prediction_score" for every row,

which represents the statistical probability that a specific title matches the

profile of a top-tier show. The final output is a ranked list of the 10 titles

with the highest probability of being a "hit" based on their core characteristics.

Conclusion

In conclusion, this project successfully demonstrates the full data science pipeline

The dataset provided valuable insights into anime trends, highlighting both industry growth and stable audience behavior. The analysis confirmed that preprocessing and feature engineering play a crucial role in obtaining meaningful results from raw data.

Overall, the project shows that even relatively simple datasets can provide significant insights when properly processed and analyzed. The skills developed in this project are highly relevant for future work in data science and analytics.